Catering Staff Management System — Module Plan

For: Badusha and Murshid's Event Management Company (Kerala)
Purpose: break the system into modules for production development and billing.
Source: `Catering_Staff_Management_System_Project_Report-5.md`

Module Map

ID	Module	Category	Tier
S1	Roles & Permissions	Shared	MVP
S2	Authentication & Sessions	Shared	MVP
S3	Staff Profile	Shared	MVP
S4	Work Core (Data & State Engine)	Shared	MVP
S5	Notifications Engine	Shared	MVP (2 sub-features Optional)
S6	Media Storage (Aadhaar & Photos)	Shared	MVP
U1	Registration	User Portal	MVP
U2	Home / Work Feed	User Portal	MVP
U3	Apply to Work	User Portal	MVP
U4	My Works	User Portal	MVP
U5	Attendance Marking	User Portal	MVP
U6	Promotion Request	User Portal	MVP
U7	Notifications Inbox	User Portal	MVP
U8	Profile	User Portal	MVP
A1	Verification Queue	Admin Panel	MVP
A2	Work Management	Admin Panel	MVP
A3	Promotion Requests Queue	Admin Panel	MVP
A4	Manual Role Management	Admin Panel	MVP
A5	Staff Directory & Filtering	Admin Panel	MVP
A6	Notification Broadcast Tools	Admin Panel	MVP (Custom Messages Optional)
A7	Account & Device Security	Admin Panel	MVP
A8	Sub Admin Oversight	Admin Panel	MVP (kept lightweight)
F1	Payment Gateway & Financial Tracking	Future	Deferred
F2	Radius/Circle Search	Future	Deferred

Most of v1 is MVP by design — the scope was already trimmed hard during planning. Only two small pieces are genuinely optional (below), and two are explicitly future.

Shared Modules

Used by both portals, or foundational to everything else.

S1. Roles & Permissions — `MVP`

- **Purpose:** single definition of every role and what it's allowed to do.
- **Features:** 10-role table (Main Admin, Sub Admin, Office Staff, D/C/B/A Boy, Supervisor, Captain, Site Manager); permission checks used by every other module; promotion state machine (automatic D→C→B, request-based B→A, manual above).
- **Roles:** all.
- **Reusable:** yes — every permission check in the system reads from here. Nothing should hardcode a role name elsewhere.

S2. Authentication & Sessions — `MVP`

- **Purpose:** get people in securely, differently by side.
- **Features:** OTP send/verify for field roles (D Boy → Site Manager); password + device-lock for admin roles (Sub Admin, Office Staff); session/token issuance.
- **Roles:** all.
- **Reusable:** yes — one auth backend, two entry flows.

S3. Staff Profile — `MVP`

- **Purpose:** the core person record.
- **Features:** personal info, Aadhaar reference, emergency contact, location (optional), position, Total Attended Works, Total Late Work, Total Late Minutes.
- **Roles:** all (self); Office Staff/Sub Admin/Main Admin (others').
- **Reusable:** yes — read by Directory, Verification, Dashboards, name cards everywhere.

S4. Work Core (Data & State Engine) — `MVP`

- **Purpose:** the Work (event) model and its rules — not the screens, the engine underneath them.
- **Features:** status machine (Created → Coming → Ongoing → Finished); application status (Confirmed/Substitute/Full); same-day rule + office-override exception; attendance → promotion-count linkage.
- **Roles:** consumed by U2–U6 and A2–A3.
- **Reusable:** yes — Work Management (admin) and Work Feed/Apply (portal) both sit on this one engine, so status logic only exists once.

S5. Notifications Engine — `MVP` (Reminder and Custom Message are `Optional`)

- **Purpose:** one place that sends push and SMS, regardless of who triggers it.
- **Features:** FCM push (automatic), MSG91 SMS (manual, District → PIN wave staging); 7 types — verification result, new work posted, work edited/cancelled, reminder (night before), custom message, position upgrade, ban.
- **Roles:** triggered by S1/S4/A1/A2/A6; received by all.
- **Reusable:** yes — every other module calls this one engine instead of sending anything itself.
- **Note:** Reminder needs a daily scheduled job; Custom Message needs a small compose UI in A6. Both are small, separable units — fine to cut from v1 if timeline is tight.

S6. Media Storage (Aadhaar & Photos) — `MVP`

- **Purpose:** upload, store, and (eventually) expire sensitive files.
- **Features:** Cloudinary upload for Aadhaar front/back + profile photo; retention policy hook (policy itself still open — see Questions/Risks).
- **Roles:** all (upload own); Office Staff/Sub Admin/Main Admin (view for verification).
- **Reusable:** yes — same upload component for registration and later profile-photo edits.

User Portal Modules

The PWA used by every field role, D Boy through Site Manager. Same app; features gate by role.

U1. Registration — `MVP`

- **Purpose:** get a new worker into the system.
- **Features:** personal info, optional location capture, profile photo (completable later), Aadhaar front/back, emergency contact + relation, 4 consent checkboxes, OTP verify.
- **Roles:** anyone not yet registered.
- **Reusable:** no — one-time flow, though its upload component reuses S6.

U2. Home / Work Feed — `MVP`

- **Purpose:** see what work is available.
- **Features:** listed work, District/PIN filters, open-to-all by default.
- **Roles:** D/C/B/A Boy (apply); Supervisor/Captain/Site Manager (view only, no apply button).
- **Reusable:** the work-card component reused in U4 and A2.

U3. Apply to Work — `MVP`

- **Purpose:** self-register for a listed work.
- **Features:** Confirmed/Substitute/Full assignment; same-day block (S4); cancellation via office call, not self-service.
- **Roles:** D/C/B/A Boy only.
- **Reusable:** no, but shares S4's logic entirely.

U4. My Works — `MVP`

- **Purpose:** track what you're registered or assigned to.
- **Features:** status per work (confirmed/substitute), history.
- **Roles:** all field roles.
- **Reusable:** work-card component shared with U2.

U5. Attendance Marking — `MVP`

- **Purpose:** mark who showed up, and handle the late-arrival call.
- **Features:** staff list for the work; 45-minute grace window; "[Name] is late?" Yes/No prompt past that; Yes writes to S3's late totals, No just records the time; same-day-override assignments skip the prompt.
- **Roles:** whoever the Site Manager (or Office Staff+) assigned to take attendance for that work — usually Captain/Supervisor.
- **Reusable:** the name-card component shared with A5 and A3.

U6. Promotion Request — `MVP`

- **Purpose:** let a Site Manager or Captain flag a standout Boy.
- **Features:** request from a work's staff list; stacks a counter (2×, 3×) on repeat requests; no cap, no deadline.
- **Roles:** Site Manager, Captain only (sending); feeds A3.
- **Reusable:** name-card component shared with U5, A5.

U7. Notifications Inbox — `MVP`

- **Purpose:** see what S5 has sent you.
- **Features:** all 7 types, read/unread.
- **Roles:** all.
- **Reusable:** no — but purely a display layer over S5, so it's thin.

U8. Profile — `MVP`

- **Purpose:** view/edit your own record.
- **Features:** edit personal info/photo/location; view position, attended/late stats (read-only, system-generated).
- **Roles:** all.
- **Reusable:** no — but reads entirely from S3.

Admin Panel Modules

Used by Office Staff, Sub Admin, Main Admin. Likely a separate, more controlled interface from the field PWA (see Questions/Risks — this decision isn't locked yet).

A1. Verification Queue — `MVP`

- **Purpose:** approve or reject new registrations.
- **Features:** review Aadhaar + details; reject with a reason, resubmit loop.
- **Roles:** Office Staff.
- **Reusable:** name-card + document-viewer components shared with A5.

A2. Work Management — `MVP`

- **Purpose:** create and run work listings.
- **Features:** create/edit/cancel/finish; optional per-tier headcount targets; office-override same-day assignment (call a running Site Manager, get names, assign directly); SMS wave trigger for a listing.
- **Roles:** Office Staff (primary), Site Manager (can also mark Finished).
- **Reusable:** work-card component shared with U2/U4.

A3. Promotion Requests Queue — `MVP`

- **Purpose:** review nominations from U6 and act on them.
- **Features:** name card + requester + request count; open profile, manually promote or ignore; no deadline.
- **Roles:** Office Staff.
- **Reusable:** name-card component shared with A5, U5.

A4. Manual Role Management — `MVP`

- **Purpose:** the one-time workforce migration, plus anything above A Boy.
- **Features:** bulk-assign existing staff to their real position at rollout; manual promote to Supervisor/Captain/Site Manager.
- **Roles:** Main Admin, Sub Admin, Office Staff.
- **Reusable:** no — mostly a rollout-week tool, though the underlying "change position" action is shared with A3.

A5. Staff Directory & Filtering — `MVP`

- **Purpose:** find and contact anyone in the system.
- **Features:** filter by District/PIN/position; tap a card for full profile.
- **Roles:** Office Staff, Sub Admin, Main Admin.
- **Reusable:** name-card + filter-bar components reused in A1, A3, U2's filters.

A6. Notification Broadcast Tools — `MVP` (Custom Message is `Optional`)

- **Purpose:** the sending side of S5 for things that need a human decision.
- **Features:** trigger an SMS wave (PIN → District → neighbouring District); send a ban notice (goes to all users); compose a custom message to a work's registered users (optional add-on).
- **Roles:** Office Staff.
- **Reusable:** no — a UI layer over S5.

A7. Account & Device Security — `MVP`

- **Purpose:** protect the accounts with access to verification and staff data.
- **Features:** first-login device approval for Sub Admin and Office Staff; block unapproved new devices; session history and termination.
- **Roles:** Main Admin (manages); Sub Admin, Office Staff (subject to it).
- **Reusable:** no — sits on top of S2.

A8. Sub Admin Oversight — `MVP`, kept lightweight

- **Purpose:** let Sub Admin see what Office Staff is doing, without giving them execution power.
- **Features:** activity overview — verifications handled, work created/finished, promotions requested/decided. No separate analytics layer for v1.
- **Roles:** Sub Admin.
- **Reusable:** no.

Future Modules

Explicitly out of v1. Don't build, even partially — but don't design v1 in a way that blocks them either.

F1. Payment Gateway & Financial Tracking — `Deferred`

- **Purpose:** profit/financial visibility for Main Admin, once the system has run for months.
- **When it comes:** additive tables only. Never modify S3/S4's existing models or migrations.

F2. Radius/Circle Search — `Deferred`

- **Purpose:** find available staff within X km of an event location.
- **Groundwork already in v1:** U1's optional location capture exists specifically so this doesn't require re-contacting every user later.

Reusable Components (Build Once, Use Everywhere)

- **Name card** (photo, name, role badge, phone, district) — U5, U6, A1, A3, A5. Build once as a single component with role-based action buttons (mark attendance / request promotion / verify / promote), not four separate versions.
- **Work card** (date, location, meal type, status, fill count) — U2, U4, A2.
- **Filter bar** (District / PIN / position dropdowns) — U2, A5.
- **File upload** (Aadhaar front/back, profile photo) — U1, U8.
- **Notifications Engine (S5)** — the one sender every other module calls into; nothing else should send push or SMS directly.

Folder Structure (Django)

```
catering_system/
├── manage.py
├── config/                  # settings, root urls, scheduled-job config
├── apps/
│   ├── accounts/             # S1, S2, S3, U8, A4, A7 — roles, profile, auth, device-lock
│   ├── registration/         # U1, A1 — registration + verification
│   ├── work/                 # S4, U2, U3, U4, U5, U6, A2, A3 — work, applications, attendance, promotions
│   ├── notifications/        # S5, U7, A6 — push + SMS engine and its UI
│   ├── directory/            # A5 — staff search/filter
│   └── media_storage/        # S6 — Aadhaar/photo upload + retention
├── templates/
│   ├── portal/               # User Portal (field roles)
│   └── admin_panel/          # Admin Panel (Office/Sub Admin/Main Admin)
├── static/
│   ├── portal/
│   └── admin_panel/
└── docs/                     # AGENTS.md, CLAUDE.md, spec/scope/decisions/progress, PROJECT_REPORT.md
```

Six apps, not more — enough to keep boundaries clean without splitting into a folder per tiny feature. `payments` and `radius_search` don't exist yet; when F1/F2 are greenlit, they're added as new apps, never merged into existing ones.

Development Order

1. **Foundation** — S1, S2, S3, S4 (schema only), S6 (upload plumbing). Build this together, not split — everything else depends on it staying consistent.
2. **Registration loop** — U1, A1, S5 (verification-result type). First end-to-end loop: someone signs up, gets approved.
3. **Work loop** — A2, U2, U3, U4. Second end-to-end loop: work gets posted, people apply.
4. **Attendance & promotions** — U5, S4 (status transitions), U6, A3, A4.
5. **Remaining notifications & directory** — rest of S5's types, A5, A6, U7.
6. **Security & oversight** — A7, A8, U8's stats display.
7. **Hardening & launch prep** — end-to-end QA; Aadhaar retention decision finalized; registration-day scaling checklist from the report (Section 12) — not code, but a go-live gate.

Team Split for 2 Programmers

Build the Foundation phase together — splitting shared models between two people early is how role/status names drift, which is exactly what `docs/spec.md` exists to prevent.

After that, split by portal, not by feature:

- **Programmer A — User Portal:** U1–U8, end to end. Owns the field-facing PWA.
- **Programmer B — Admin Panel:** A1–A8, end to end. Owns the office-facing side.
- **Shared modules (S1–S6):** owned by whoever built them in Foundation. Both programmers consume them through their existing functions — neither edits the other's app directly. If a shared module needs to change mid-build, that's a conversation between both,

not a solo edit.

This keeps the two programmers in different folders (`apps/work`'s portal-facing views vs. its admin-facing views can even be split within the same app if needed), which keeps merge conflicts rare and makes it obvious whose work a bug belongs to.

Questions / Risks

1. **Admin Panel build approach isn't decided.** A customized Django admin is far cheaper than a fully separate dashboard UI, and this changes every A-module's estimate. Worth deciding before quoting, not after.
2. **This plan assumes Supervisor/Captain/Site Manager use the same PWA as D/C/B/A Boy**, just with extra role-gated actions (U5, U6). That's an assumption, not something the report states outright — flag it if that's wrong, since it would move U5/U6 into a different app entirely.
3. **DB engine is still unconfirmed** (Postgres recommended in `conventions.md`, not yet decided). Affects hosting setup timing, not the module plan itself.
4. **Reminder notifications need a daily scheduled job.** Recommend a simple cron + management command over standing up Celery/Redis — that's real infrastructure for what's fundamentally one batch send a day. Flag if there's a reason to want the heavier setup.
5. **Aadhaar retention policy is still open** (per Section 14 of the report). It doesn't block building S6's upload/verify flow, but the auto-delete job itself is a separate, small piece of work once the policy is picked.